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Computational biodynamics of human knee joint in gait: From muscle forces to cartilage stresses

M. Adouni [a], A. Shirazi-Adl [a,*], R. Shirazi [b]

[a] Division of Applied Mechanics, Department of Mechanical Engineering, École Polytechnique, P.O. Box 6079, Station "centre-ville", Montréal, Québec, H3C 3A7 Canada
[b] Department of Bioengineering and Whitaker Institute of Biomedical Engineering, University of California—San Diego, San Diego, California

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ABSTRACT

Using a validated finite element model of the intact knee joint we aim to compute muscle forces and joint response in the stance phase of gait. The model is driven by reported *in vivo* kinematics–kinetics data and ground reaction forces in asymptomatic subjects. Cartilage layers and menisci are simulated as depth-dependent tissues with collagen fibril networks. A simplified model with less refined mesh and isotropic depth-independent cartilage is also considered to investigate the effect of model accuracy on results. Muscle forces and joint detailed response are computed following an iterative procedure yielding results that satisfy kinematics/kinetics constraints while accounting at deformed configurations for muscle forces and passive properties. Predictions confirm that muscle forces and joint response alter substantially during the stance phase and that a simplified joint model may accurately be used to estimate muscle forces but not necessarily contact forces/areas, tissue stresses/strains, and ligament forces. Predictions are in general agreement with results of earlier studies. Performing the analyses at 6 periods from beginning to the end (0%, 5%, 25%, 50%, 75% and 100%), hamstrings forces peaked at 5%, quadriceps forces at 25% whereas gastrocnemius forces at 75%. ACL Force reached its maximum of 343 N at 25% and decreased thereafter. Contact forces reached maximum at 5%, 25% and 75% periods with the medial compartment carrying a major portion of load and experiencing larger relative movements and cartilage strains. Much smaller contact stresses were computed at the patellofemoral joint. This novel iterative kinematics-driven model is promising for the joint analysis in altered conditions.

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1. Introduction

Activities of daily living such as walking and stair climbing impose relatively large loads and movements on the human knee joint. This mechanical burden increases in many occupational and recreational tasks causing injuries and degenerations in joint ligaments, menisci, cartilage and bone. Any failure, degeneration or alteration in one of these components influences the response of the entire joint and likely increases the risk of further perturbations (Moglo and Shirazi-Adl, 2005). Effective preventive and conservative/surgical managements of joint disorders depend hence on a sound knowledge of stress and strain distributions in various components under both intact and altered conditions. These values, in turn, are heavily dependent not only on external loads and inertial forces but on muscle activities across the joint. As such, accuracy in estimation of muscle forces has a direct bearing on the reliability of stresses and strains. Since direct *in vivo* measurements of tissue stresses and muscle forces remain invasive and likely impossible, computational modeling is recognized as a vital complementary tool to estimate multiple variables of interest. Due to technical difficulties in measurements and consideration of physiological loads and motions, *in vitro* testing is also limited especially when looking for cartilage/meniscus stresses/strains and ligament forces.

Using instrumented knee implants, efforts have been directed to *in vivo* measurement of loads on the knee joint of subjects in daily activities (Kutzner et al., 2010). Forces up to ~3 times body weight have been recorded in walking. These measured data, though very valuable, are however collected on implanted joints and not intact ones. Moreover, they are restricted to the load components transmitted via the implant itself. Mathematical modeling along with joint kinematics and ground reaction forces are combined with such measurements to estimate muscle, compartmental, and ligament forces (Anderson and Pandy, 2001; Delp et al., 2007; Kim et al., 2009; Lin et al., 2010; Shelburne et al., 2004; Zhao et al., 2007). To gain further insight into biomechanics of the joint during gait, some recent works have employed MRI, video motion systems and fluoroscopy to

* Corresponding author: Fax: +1 514 340 4176.
E-mail address: abshir@meca.polymtl.ca (A. Shirazi-Adl).

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estimate changes in ACL length, contact areas and cartilage thickness (Andriacchi et al., 2009; Coleman et al., 2011; Koo et al., 2011; Taylor et al., 2011) or to identify the effect of ACL-deficiency on joint kinematics (Chen et al., 2011). The medial plateau is found to experience larger tibiofemoral movements and cartilage strains.

In parallel, other studies have used 3D link-segment models (inverse dynamics and optimization) to predict load distribution between medial and lateral compartments as well as ligament forces during simulated gait (Shelburne et al., 2004, 2005, 2006). These models, however, explicitly neglect not only the compliant cartilage layers and menisci but the knee joint passive resistance. Similarly and using measured *in vivo* load in one subject with instrumented knee, joint contact forces have been estimated during the gait (Lin et al., 2010; Zhao et al., 2007). Simulating cartilage and menisci with linear elastic properties, a recent finite element (FE) model study in gait has on the other hand neglected joint rotations, redundancy in equations, patellofemoral joint, and out-of-sagittal plane moments (Yang et al., 2010).

Despite foregoing *in vivo* measurement and model studies, a detailed musculoskeletal FE model of the entire intact knee joint with depth-dependent nonlinear cartilage/meniscus properties remains yet to be developed. This model study should provide important data during gait on not only the muscle forces but the ligament forces, contact stresses as well as stresses/strains within the cartilage and menisci. Such is the aim of the current study that is based directly on earlier developments, validations and applications of a detailed FE model of the knee joint (Adouni and Shirazi-Adl, 2009; Bendjaballah et al., 1995; Mesfar and Shirazi-Adl, 2006; Moglo and Shirazi-Adl, 2005; Shirazi et al., 2008). In the current study, cartilage of both tibiofemoral (TF) and patello-femoral (PF) joints along with menisci are simulated as depth-dependent non-homogeneous composites enclosing nonlinear collagen fibril networks (Shirazi and Shirazi-Adl, 2009a,b; Shirazi et al., 2008). For the sake of comparison, a less-refined model with isotropic (no fibril networks) depth-independent cartilage is also considered. Both models incorporate the mean reported *in vivo* gait data on hip/knee/ankle joint moments/rotations (Astephen et al., 2008a) and ground reaction forces (Hunt et al., 2001) of asymptomatic subjects. In this manner, biodynamics of gait during the stance phase, at both global musculature and local tissue-level, are investigated with iterative FE models of the knee joint and lower extremity that are driven by measured *in vivo* kinetics/kinematics. We hypothesize that (1) muscle activation levels and contact stresses/areas substantially alter during the stance phase and (2) in contrast to contact stresses/areas, ligament forces and tissue stresses/strains, the muscle forces can as accurately be estimated by the less complex FE model.

2. Methods

2.1. Finite elements model

The FE model includes bony structures (tibia, patella and femur), TF and PF joints, major TF (ACL, PCL, LCL and MCL) and PF (MPFL and LPFL) ligaments, patellar tendon (PT), as well as quadriceps (3 distinct muscles), hamstrings (3 muscles), gastrocnemius (2 muscles), tibialis posterior and soleus (Adouni and Shirazi-Adl, 2009; Shirazi et al., 2008; see Fig. 1 for details). The bony structures are represented by rigid bodies due to their much higher stiffness (Donahue and Hull, 2002).

Muscle components are modeled by uniaxial elements with orientations at full extension taken from the literature. The Q angle model ($Q=14^\circ$; Sakai et al., 1996) is used for quadriceps muscles; orientations relative to the femoral axis in frontal/sagittal planes are: RF–VIM $0^\circ/4^\circ$ anteriorly, VL 22° laterally/0° and VMO 41° medially/0°. Orientations for hamstrings muscles relative to the tibial axis, respectively for BF, SM, and TRIPOD are taken (Aalbersberg et al., 2005) as 11.8° medially, 7° laterally, and 7.1° medially in the frontal plane whereas 0°, 16.1°, and 18.7° posteriorly in the sagittal plane. Gastrocnemius fascicles are parallel to the tibial axis in the sagittal plane while oriented (GM) 5.3° medially or (GL) 4.8° laterally in the frontal plane (Delp et al., 2007; Hillman, 2003). Tibialis posterior/Soleus are oriented $5.3^\circ/4.1^\circ$ laterally and $1.0^\circ/4^\circ$ anteriorly relative to the tibial

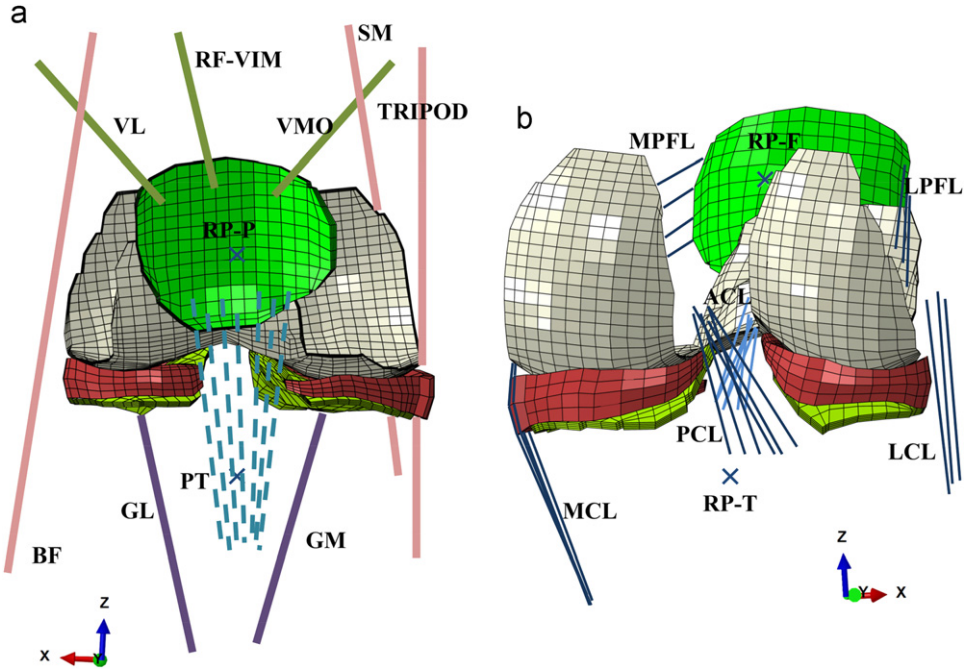


Fig. 1. (a) Knee joint FE model; tibiofemoral (TF) and patellofemoral (PF) cartilage layers, menisci, patellar tendon (PT), quadriceps, hamstrings and gastrocnemius muscles, (b) TF and PF cartilage layers, menisci and joint ligaments. Crosses indicate the reference points (RP) representing rigid bony structures that are not shown. Quadriceps components are vastus medialis obliqus (VMO), rectus femoris (RF), vastus intermidus medialis (VIM), and vastus lateralis (VL). Hamstring components include biceps femoris (BF), semimembranous (SM), and the TRIPOD made of sartorius (SR), gracilis (GR), and semitendinosus (ST). Gastrocnemius components are gastrocnemius medial (GM) and gastrocnemius lateral (GL). Tibialis posterior and soleus muscles of the ankle joint are not shown. Joint ligaments include lateral patellofemoral (LPFL), medial patellofemoral (MPFL), anterior cruciate (ACL), posterior cruciate (PCL), lateral collateral (LCL), and medial collateral (MCL).

axis (Delp et al., 2007). Ligaments are each modeled by a number of uniaxial elements with different initial pre-strains, non-linear (tension-only) material properties, and initial cross-sectional areas of 42, 60, 18, 25, 99, 42.7 and 28.5 mm^2 for ACL, PCL, LCL, MCL, PT, MPFL, and LPFL, respectively (Mesfar and Shirazi-Adl, 2005).

Articular cartilage layers and menisci are modeled as depth-dependent composites of an isotropic bulk reinforced by networks of collagen fibrils. In the cartilage superficial zone, fibrils are oriented horizontally parallel to the surface whereas they become random in the transitional zone and then turn perpendicular in the deep zone anchoring into the subchondral bone. Membrane elements are used to simulate fibril networks in the superficial and deep zones while brick elements represent the transitional zone network. In menisci, collagen fibrils are primarily oriented in the circumferential direction within the bulk but have no preferred orientation on bounding surfaces (Shirazi and Shirazi-Adl, 2009a; Shirazi et al., 2008).

2.2. Material properties

Depth-dependent isotropic hyperelastic (Ogden-Compressible) material properties are considered for non-fibrillar solid matrix of cartilage layers with the elastic modulus varying linearly from 10 MPa at the surface to 18 MPa at the deep zone and a Poisson's ratio of 0.49. In the short-term stance phase loading, this model with a compressible material is initially verified by additional simulations to be equivalent to the incompressible elastic model with much smaller moduli (in the range 0.3–1.2 MPa) used earlier (Shirazi and Shirazi-Adl, 2009a; Shirazi et al., 2008). Compressible hyperelastic model was initially also employed for the non-fibrillar menisci but due to convergence problems at contact areas, the matrix of menisci was represented, similar to our earlier studies (Mesfar and Shirazi-Adl, 2005), by a compressible elastic material with a Young's modulus of 10 MPa and a Poisson's ratio of 0.45.

For cartilage collagen fibrils volume fractions, 15% is considered in the superficial region, 18% in the transitional region, and 21% in the deep zone. Thicknesses of these zones are 15%, 22.5%, and 62.5% of the total value at each point, respectively. In the menisci, the collagen content is 14% in the circumferential direction and 2.5% in the radial direction of the bulk region along with 12% in the outer surfaces at both directions (Shirazi and Shirazi-Adl, 2009b; Shirazi et al., 2008).

For the sake of comparison with this depth-dependent fibril reinforced model, another less-refined model with isotopic representation of cartilage layers, similar to the earlier generation of our model (Bendjaballah et al., 1995; Mesfar and Shirazi-Adl, 2006, 2008), is also used. The differences are limited to the less refinement of cartilage/menisci meshes as well as the depth-independent isotropic (modulus of 12 MPa and Poisson's ratio of 0.45) representation of cartilage layers.

2.3. Muscle force estimation

An optimization technique is used to evaluate unknown muscle forces at each instance of stance phase. Cost function of the sum of cubed muscle stresses (Eq. (1); Arjmand and Shirazi-Adl, 2006) is used along with inequality equations on muscle forces remaining positive and larger than their passive forces but smaller than the sum of their passive and maximum active forces (Eq. (3)). The passive forces are neglected due to the expected negligible increases in muscle lengths while the maximum active stress is taken as 0.6 MPa (Arjmand and Shirazi-Adl, 2006). Prescribed joint rotations generate 3 moment equilibrium equations that act as constraints (Eq. (2)).

$$\text{Cost function} = \sum_{i=1}^{n} \left(\frac{F_i}{PCSA_i}\right)^3 \tag{1}$$

$$\sum_{i=1}^{n} r_{ij} F_{ij} = M_{j1,2,3} \tag{2}$$

$$F_{pi} \le F_i \le (F_{pi} + \sigma_{i\max} PCSA_i) \tag{3}$$

with F_i,F_{pi},r_{ij},$\sigma_{i\max}$ and $PCSA_i$ being the force, passive force component, lever arms in different planes, maximum stress, and physiological cross-sectional areas of muscle i (Delp et al., 2007), respectively. M_j are required moments computed in the FE model under prescribed rotations.

2.4. Loading, kinematics and boundary conditions

The hip/knee/ankle joint rotations/moments and ground reaction forces at foot during the stance phase (Fig. 2) are taken from the mean data of *in vivo* measurements on asymptomatic subjects during gait (Astephen et al., 2008a; Hunt et al., 2001). The location of resultant ground reaction force at each instant is determined so as to generate reported joint moments (Astephen et al., 2008a) accounting for the leg/foot weight (29.78 N/7.98 N). Non-orthogonal local joint coordinate systems (Grood and Suntay, 1983) are considered in compliance with prescribed rotations (Astephen et al., 2008a). Since our model was constructed

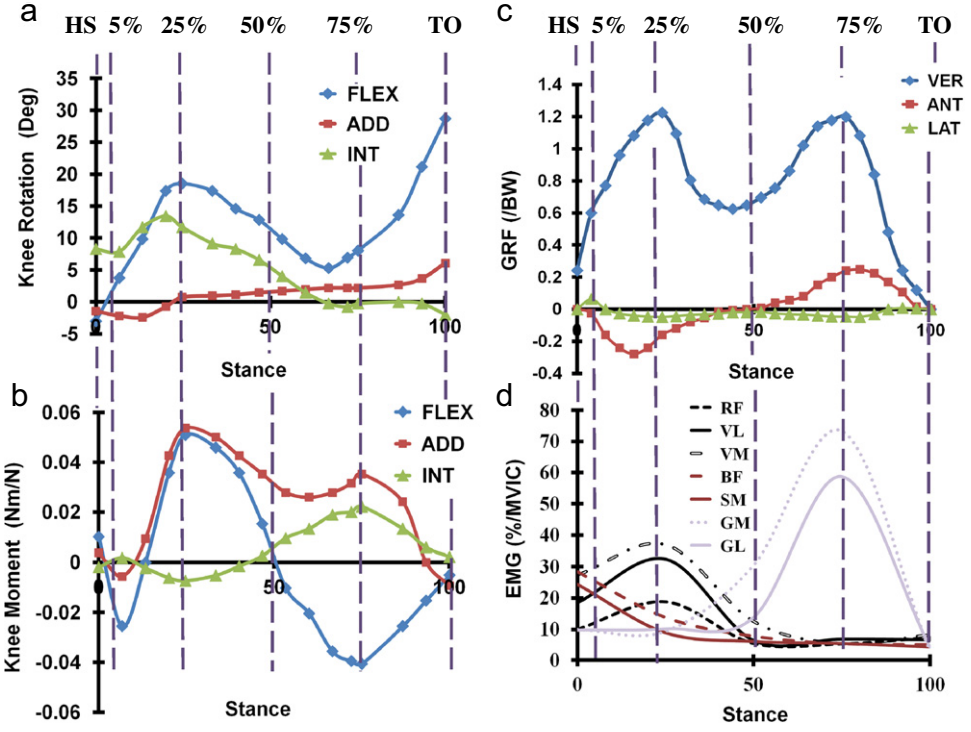


Fig. 2. (a) Knee joint rotations and (b) moments reported as mean of asymptomatic subjects during the stance phase of gait (Astephen et al., 2008a) as well as (c) vertical, anterior, and lateral ground reaction force components reported during the stance phase of gait on asymptomatic subjects (Hunt et al., 2001), and (d) mean normalized surface EMG data recorded in 7 different knee joint muscles (Astephen, 2007). For muscle abbreviations, see the caption of Fig. 1. Six instances corresponding to beginning (heel strike, HS 0%), 5%, 25%, 50% and 75% and end (toe off, TO 100%) of the stance phase are indicated. Note that moments and forces are normalized to the BW=606.6 N of the female subject.

based on a female knee joint, a body weight of BW=606.6 N (61.9 kg) is considered (De Leva, 1996). Analyses are carried out at 6 time instances corresponding to HS (heel strike), 5%, 25%, 50% and 75%, and TO (toe-off) of stance phase (Fig. 2). At each period the femur is initially fixed in its instantaneous position while the tibia and patella are completely free except for the prescribed TF rotations.

At each stance period under associated prescribed TF rotations (Fig. 2), ground reaction forces (Fig. 2), leg/foot weight, and GL/GM forces (already evaluated via equilibrium of moments at the ankle joint alongside tibialis posterior and soleus muscles), unknown forces in remaining muscles are iteratively estimated. This is done by initially evaluating muscle forces (Eqs. (1)–(3)) counterbalancing required moments in the deformed configuration. These forces are subsequently applied in the model as additional external loads and the procedure is repeated (8–12 iterations) till convergence is reached, when unbalanced required moments fall below 0.1 Nm. In this manner, both kinematics and kinetics conditions are simultaneously satisfied while accounting for the penalty of muscle forces and the joint passive resistance. Matlab (R2009a Optimization Toolbox, genetic algorithms) and ABAQUS 6.10.1 (Static analysis) commercial programs are used iteratively at each phase.

3. Results

Muscle forces in GL and GM initiated at 25% and reached their maximum, respectively of 0.29 BW and 0.88 BW, at 75% period (Fig. 3). Quadriceps forces peaked at 25% (Fig. 3) with the RF–VI carrying the largest portion at 0.74 BW compared to 0.58 BW in VL and 0.40 BW in VM. The PT force followed similar trend reaching maximum of 1.56 BW at 25% (Fig. 3). While resisting the adduction moment (Fig. 2), BF muscle peaked at 1.00 BW at 5% (Fig. 3). Almost the same muscle forces were computed in the simplified (isotropic cartilage) model (max difference of 0.02 BW).

Among ligaments, ACL force was the largest followed by the LCL and MCL while PCL remained unloaded throughout and PF ligaments resisted small forces (each <20 N). ACL Force (PL bundle only) increased from 231 N at HS to its peak of 343 N at 25% and decreased thereafter (Fig. 4). The MCL was loaded only at HS and 5% period while in contrast the LCL was primarily loaded at TO (Fig. 4). Use of the simplified model markedly influenced ligament forces; e.g., peak ACL force dropped to 275 N at 25% while peak LCL force increased to 238 N at TO (Fig. 4).

Except at the HS and 5% period, the medial plateau carried much larger forces than the lateral one with peaks at 25% and 75% periods (Fig. 5). The load was transmitted primarily through the uncovered areas (via cartilage–cartilage; Fig. 5). The TF contact area reached also its maximum at 25% (922 mm^2) and 75% (1017 mm^2) periods (Fig. 6). The PF contact force and area followed the same trend as quadriceps forces and reached their peak at 25% period; 459 N and 232 mm^2. Large differences were computed when using the less refined model with isotropic cartilage; TF contact forces were smaller (by 5.4% at 25%) while contact areas were larger (by 168 mm^2 or 18.2% at 25% and 206 mm^2 or 20.2% at 75%; Fig. 6).

In accordance with the compartmental loads (Fig. 5), contact pressures were much larger on the medial plateau where they shifted posteromedially during the stance and reached their peak of 8.1 MPa at 25% period (Fig. 7). Tensile strains were also larger in the cartilage of the medial plateau reaching greater values at deep fibril networks (peak of 16% at 25% period; Fig. 8). The simplified isotropic model computed smaller contact pressures at TF and PF contact areas (by 19% for mean pressure and 30% for peak pressure).

4. Discussion

A novel passive–active FE model of the lower extremity is developed to predict the global–local biodynamics of the knee joint during stance phase of gait. The model accounts for major muscle groups crossing the joint and represents the cartilage and menisci as depth-dependent nonlinear non-homogeneous composites with collagen fibril networks. It is driven by *in vivo* measured hip/knee/ankle kinematics (Astephen et al., 2008a) and ground reaction forces (Hunt et al., 2001) reported for asymptomatic subjects. Muscle forces and joint detailed response are computed following an iterative procedure yielding results that satisfied kinematics/kinetics constraints while accounting at deformed configurations for the muscle forces as well as nonlinear passive properties. A simplified model of the joint with less refined mesh and isotropic depth-independent cartilage layers is also considered to investigate the effect of model accuracy on results. Predictions confirm the hypotheses in that (1) muscle forces and detailed joint response alter substantially during the stance phase and (2) a simplified joint model may accurately be

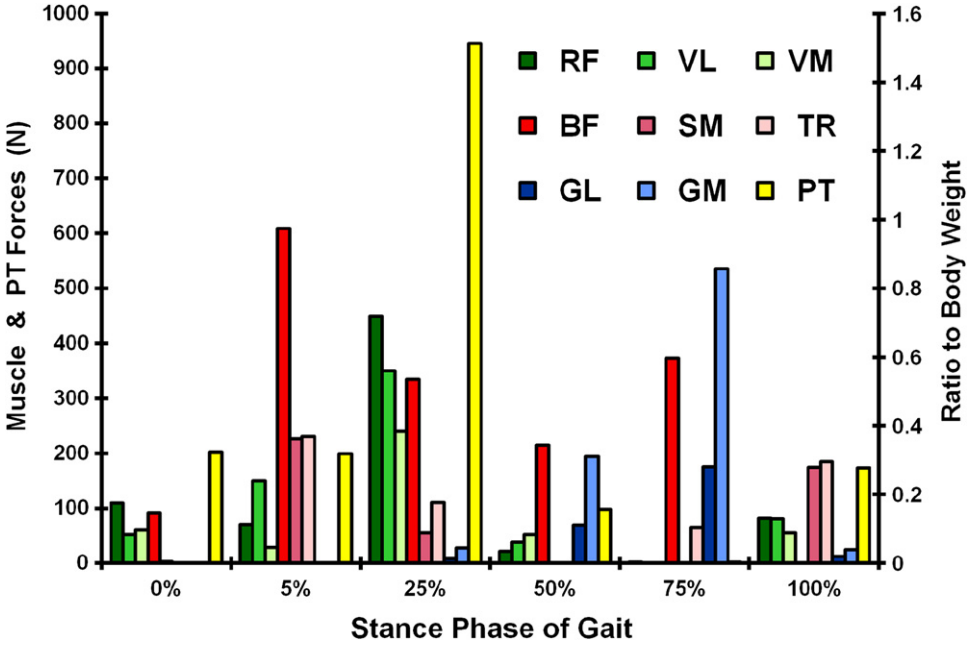


Fig. 3. Predicted muscle and TP forces at different stance phases (see Fig. 1 caption for muscle abbreviations). Ratios to body weight are given on the right axis.

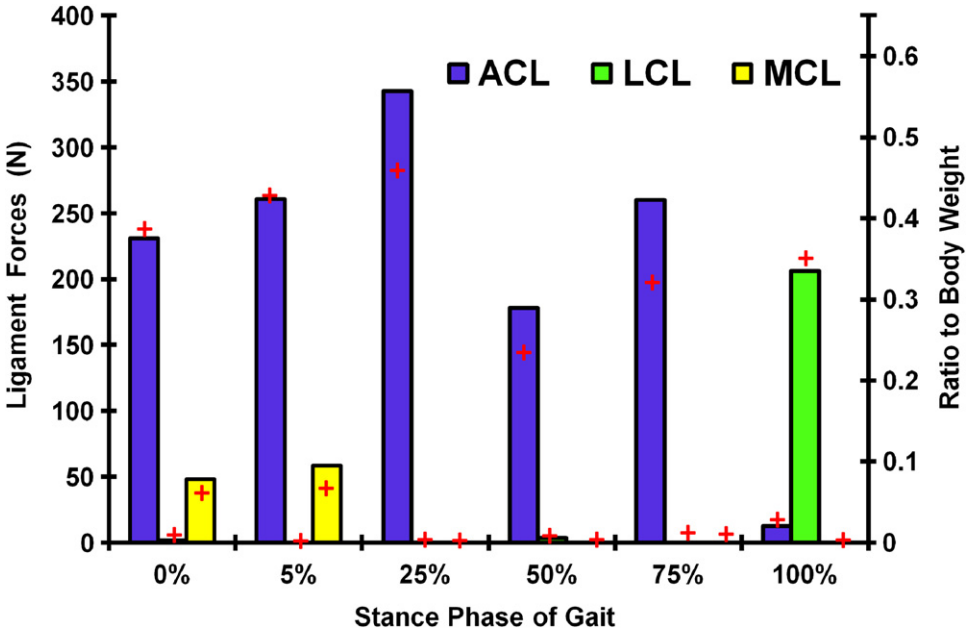


Fig. 4. Predicted ligaments forces at different stance phases (see Fig. 1 caption for ligaments abbreviations). (+) Results found in the model with less-refined mesh and isotopic representation of cartilage layers. Ratios to body weight are given on the right axis.

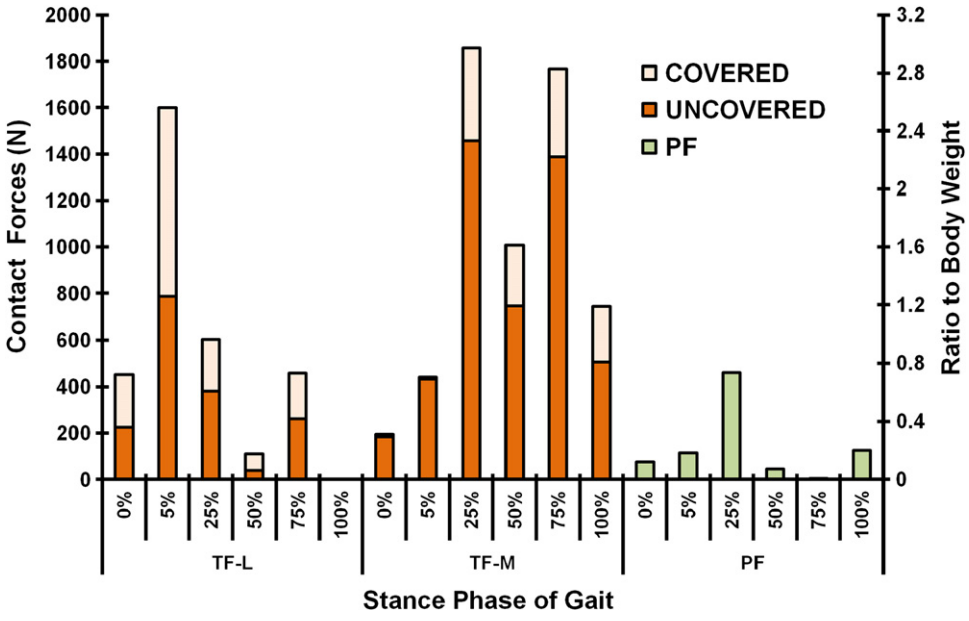


Fig. 5. Predicted total TF (covered: via menisci, uncovered: via cartilage, M: medial plateau, L: lateral) and PF contact forces at different stance phases. Ratios to body weight are given on the right axis.

used to estimate muscle forces but not necessarily contact forces/areas, tissue stresses/strains, and ligament forces.

In accordance with our earlier investigations (Shirazi et al., 2008), the transient response of the cartilage and menisci under higher strain rates during the gait can accurately be captured either by a biphasic analysis or equivalently by an incompressible elastic analysis using similar equilibrium moduli. In this study, however and due to convergence difficulties, a compressible model with much larger elastic moduli varying linearly from 10 MPa at the surface to 18 MPa at the deep zone and a Poisson's ratio of 0.49 are employed. These values were chosen after extensive preliminary parametric studies on the entire joint in axial compression while matching results of the model with incompressible materials for soft tissues (Shirazi et al., 2008).

Muscle forces in this work are estimated while satisfying the moment equations at both the ankle and knee joints. The knee joint muscles that cross the hip, however, should also participate in similar equations at the hip joint. To assess the extent of changes, had the hip joint moment equations been considered simultaneously alongside those at the knee and ankle joints, the model was extended to account for the hip joint muscles (Delp et al., 2007) and moments (Astephen et al., 2008a). Preliminary results so far (not presented here) have shown relatively small changes ($<10\%$) in estimated muscle forces when the hip joint is included. Presentation and discussion of these results will be the subject of a future work.

Earlier model studies of the knee joint during walking have neglected the joint passive resistance when computing muscle

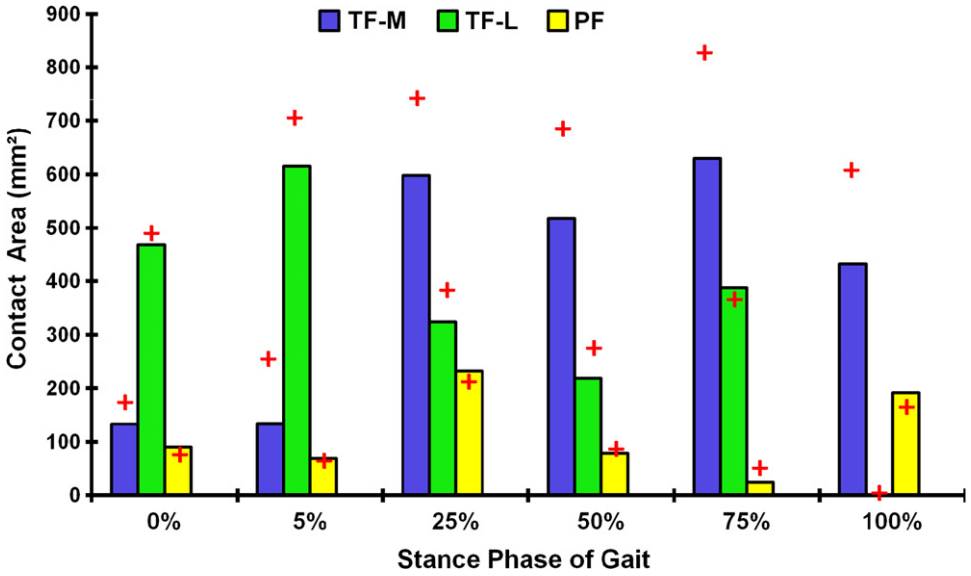


Fig. 6. Predicted total TF (M: Medial and L: Lateral) and PF contact areas at different stance phases. (+) Results found in the model with less-refined mesh and isotopic representation of cartilage layers.

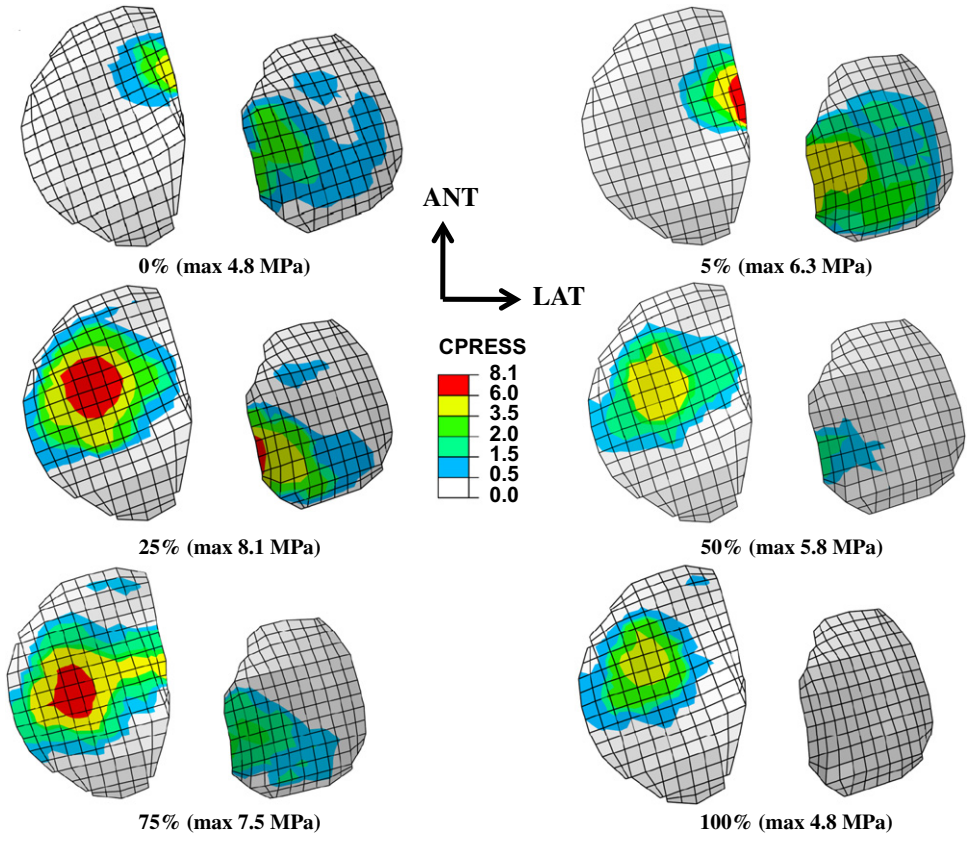


Fig. 7. Predicted contact pressure at articular surface of lateral and medial tibial plateaus at different stance phases. Note that a common legend is used for ease in comparisons.

forces (Shelburne et al., 2005, 2006). Ligaments, menisci and articular surfaces contribute to the passive moment carrying capacity of the joint that is expected to increase in compression. These passive moments tend to support a portion of net external moments and thus reduce required muscle forces. Precise quantification of the joint passive resistance under various compression forces needs however separate studies that are currently underway. Differences between the applied moments (i.e., inverse dynamics) and moments resisted by muscles at the final converged solution indicate passive joint flexion, adduction and internal moments of, respectively, 6.5, 9.0, and 2.1 Nm at 25% period and 5.2, 10.8, and 2.0 Nm at 75% period of stance phase.

Comparison of results when using a simplified model with less refined mesh and isotropic homogeneous cartilage layers instead of a refined mesh with depth-dependent cartilage properties and collagen fibrils networks demonstrate negligible changes in

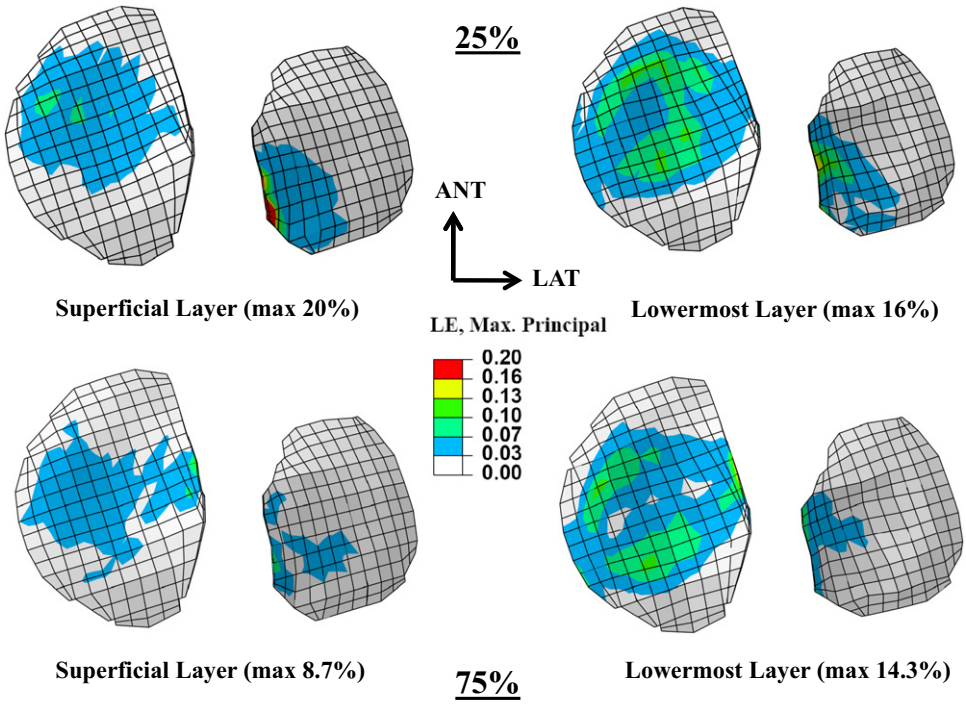


Fig. 8. Maximum (tensile) principal strain in solid matrix at the superficial and lowermost layers of lateral and medial tibial cartilage at 25% and 75% stance phases. Note that a common legend is used for ease in comparisons.

estimated muscle forces (<0.02 BW). Forces in ligaments and contact forces/stresses/areas are however altered. The contact stresses and areas are naturally affected due both to the refinement and more realistic presentation of cartilage itself. Use of the simplified model is hence justified when looking for the overall response and muscle forces. Future simulations of joint disorders (i.e., ligament injuries, cartilage defects and degenerations) justify also the use of the detailed model of the joint. It is to be emphasized that alterations in the input kinematics/kinetics, material and structural properties as well as joint flexion axis considered in this study likely influence the results the extent of which can only be quantified by proper sensitivity analyses (Dhaher et al., 2010; Moglo and Shirazi-Adl, 2005).

Predicted activation levels in quadriceps and gastrocnemius muscles are in satisfactory agreement with values in the literature (Besier et al., 2009; Lin et al., 2010; Neptune et al., 2004; Shelburne et al., 2005; Winby et al., 2009) and follow the same relative trends as measured EMG activities (Astephen, 2007; Astephen et al., 2008b). The computed hamstring forces peaked right after the HS at 5% period (Fig. 3). Reported normalized EMG activities in the superficial lateral and medial hamstrings (Astephen, 2007) are also highest right at the HS and decrease thereafter (Fig. 2(d)). Additional FE analyses at 75% stance period demonstrate that had we totally neglected the soleus muscle contribution to resist flexion at the ankle joint, we would have a substantial drop in BF activity (from 381 N to 181 N) at the expense of significant increases in GL and GM forces (from 710 N to 1512 N) as well as in quadriceps forces (from 4 N to 299 N). Such drop in BF force appears to further improve agreement with reported negligible hamstrings EMG in the 2nd phase of stance (Fig. 2(d)). Comparisons between estimated muscle forces and recorded EMG data should however also account for the (a) absence of coactivity in the model in order to enhance joint stability and control, (b) lack of consideration for hip moment equations when computing knee muscle forces, and (c) inherent limitation of select surface electrodes in measuring EMG data especially of deeper hamstrings muscles (e.g., BF short head).

As a consequence of relatively small flexion angles during the stance, the quadriceps muscle forces remain nearly equal to PT forces with a ratio of $\sim$0.95. These PT forces, in turn and due to their anterior orientation, pull the tibia anteriorly that increases ACL force (Fig. 4). With the drop in PT force, the anterior shear force and hence ACL force decrease at the later periods of the stance phase. Bursts of activity in GM and GL, especially at 75% stance period, tend to markedly decrease hamstrings forces and as a result increase ACL forces. Similar variations have been reported in earlier model studies (Shelburne et al., 2004, 2005, 2006). As expected and under these conditions, PCL remain slack. The large force in the LCL at TO period is due to the joint adduction angle and activities in medial hamstrings in response to the joint small abduction moment (Fig. 2).

Due to the adduction moment on the joint, the medial compartment carries the major portion of compression at all instances except the HS and 5% period (Fig. 5). These forces, at both plateaus, are transmitted primarily at uncovered areas via cartilage–cartilage interfaces. The disproportionate partition of load between TF compartments is corroborated by earlier works (Andriacchi et al., 2009; Hurwitz et al., 1998; Kutzner et al., 2010; Neptune et al., 2004; Shelburne et al., 2006; Thambyah, 2007; Zhao et al., 2007). In comparison, relatively low contact forces are computed at the PF joint that is due to the small joint flexion angles during the stance phase. The gait loading has not indeed been identified as a major concern for the PF joint (Mason et al., 2008).

Contact stresses as well as cartilage stresses/strains are influenced by compartmental loads. Cartilage on the medial compartment experiences greater compressive stresses/strains as well as a posterior shift in the contact area (Figs. 7 and 8). In corroboration, Koo et al. (2011) and Coleman et al. (2011) recorded during walking larger decreases in the cartilage thickness and greater posterior movements on the medial plateau compared with the lateral one. Noteworthy are the relatively large tensile strains computed in the deep vertical fibril networks of the medial cartilage under the contact region. These strains are even larger

than those in the superficial horizontal fibril networks at the same region. The crucial role of deep fibril networks in the load bearing of the cartilage has been demonstrated (Shirazi et al., 2008). These results may explain the frequent observation of the joint osteoarthritis at the medial compartment (Engh, 2003; Sharma et al., 2000).

In summary, the current novel iterative kinematics-driven FE model that accounts for the synergy between passive structures and active musculature of the knee joint was used to determine muscle forces and tissue stresses/strains during the gait. Future works will consider gait of symptomatic subjects with ACL injury and/or OA that should shed further light on biodynamics of the joint in normal and perturbed conditions.

Conflict of interest statement

There is no conflict of interest what-so-ever here in this study.

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